

Manufacturing Production Monitoring System

- Detailed Implementation Phases

Phase 1: Foundation (Weeks 1-4)

Week 1: Project Setup & Database Foundation

Day 1-2: Solution Architecture Setup

Task	Deliverable	Technical Details
Create layered solution structure	Complete project skeleton	API, BLL, DAL, Models, IoT projects with proper references
Configure dependency injection	DI container setup	Register DbContext, repositories, services in Program.cs
Set up Serilog logging	Structured logging configuration	Console + PostgreSQL sinks, log levels per component
Configure environment variables	appsettings.json structure	Development, Staging, Production configurations

Day 3-4: Database Implementation

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Task	Deliverable	Technical Details
Create Entity classes	All POCO classes based on schema	BaseEntity, Machine, WorkCenter, WorkOrder, ProductionEvent, DowntimeRecord, DowntimeReason
Configure DbContext	ApplicationDbContext	Fluent API configurations, relationships, indexes
Implement Identity tables	Extended IdentityUser	ApplicationUser with custom fields (FirstName, LastName, EmployeeId, PlantLocation, Shift)
Create initial migration	Migration script	Add-Migration InitialCreate with all tables
Seed default data	Data seeding logic	Roles (8 roles), Admin user, Downtime reasons (10+), WorkCenters (3), Machines (5)

Day 5: Repository Layer

Task	Deliverable	Technical Details
Generic repository pattern	IGenericRepository	CRUD operations, filtering, pagination
Machine repository	IMachineRepository	GetMachineWithDetails, GetMachinesByWorkCenter, GetActiveMachines, UpdateMachineStatus
WorkOrder repository	IWorkOrderRepository	GetByStatus, GetByDateRange, GetByMachine
Downtime repository	IDowntimeRepository	GetActiveDowntime, GetByMachineAndDateRange
Unit of Work pattern	IUnitOfWork	Transaction management, cross-repository coordination

Completion Criteria for Week 1:

- Database can be created via `Update-Database`
- All tables exist with proper foreign keys
- Seed data is present in lookup tables
- Repository methods pass basic CRUD tests

Week 2: JWT Authentication & User Management

Day 1-2: JWT Implementation

Task	Deliverable	Technical Details
Configure JWT authentication	JWT settings in appsettings	Secret key, issuer, audience, expiry (24 hours)
Token generation service	ITokenService	GenerateAccessToken, GenerateRefreshToken, ValidateToken
Refresh token mechanism	Token refresh endpoint	Store refresh tokens in database, rotation on each use
Password hashing	Password service	Leverage ASP.NET Core Identity for password management
Authentication controller	AuthController	Login, Register, RefreshToken, Logout, ChangePassword endpoints

Day 3-4: Role-Based Access Control (RBAC)

Task	Deliverable	Technical Details
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Task	Deliverable	Technical Details
Role seeding	CreateRoles() method	Administrator, PlantManager, ProductionSupervisor, QualityInspector, MaintenanceTechnician, ProductionPlanner, Operator, Viewer
Policy definitions	Authorization policies	"RequireAdmin", "RequireSupervisor", "RequireOperator" policies
Custom authorization attributes	[AuthorizeRole]	Attribute for controller/action level role checks
User management service	IUserService	CreateUser, UpdateUser, DeleteUser, GetUsersByRole, AssignRole, RemoveRole
User management controller	UserController	CRUD operations for users (Admin only), role assignment

Day 5: Security Hardening

Task	Deliverable	Technical Details
Rate limiting	Rate limiting middleware	5 attempts per minute for login, 100 requests per minute for API
CORS configuration	CORS policy	Allow specific Angular origins, restrict methods and headers
Security headers	Middleware	X-Content-Type-Options, X-Frame-Options, Referrer-Policy
Audit logging	Audit middleware	Log all authentication attempts, user creation/deletion, role changes

Completion Criteria for Week 2:

- Users can register and login
- JWT token works with Bearer authentication
- Different roles have different access levels
- Refresh token successfully renews expired tokens

Week 3: Machine Master Data Management

Day 1-2: Machine CRUD Operations

Task	Deliverable	Technical Details
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Task	Deliverable	Technical Details
Machine service	IMachineService	CreateMachine, UpdateMachine, DeleteMachine (soft delete), GetMachineById, GetAllMachines
Machine controller	MachineController	GET, POST, PUT, DELETE endpoints with proper authorization
Work center service	IWorkCenterService	CRUD for work centers, assign supervisor
Machine DTOs	MachineCreateDto, MachineUpdateDto, MachineResponseDto	Validation attributes, mapping profiles (AutoMapper)
Bulk import/export	Excel/CSV support	Import machines from Excel, export to CSV for reporting

Day 3: Machine Status Management

Task	Deliverable	Technical Details
Status tracking service	IMachineStatusService	UpdateStatus, GetStatusHistory, GetCurrentStatus
Status history table	MachineStatusHistory	Track status changes with timestamps and reasons
Real-time status endpoint	GET /api/machines/{id}/status	Returns current status and last event time
Status transition validation	Status transition rules	Validate allowed transitions (e.g., Fault → Maintenance, not Fault → Running without repair)

Day 4-5: API Documentation & Testing

Task	Deliverable	Technical Details
Swagger/OpenAPI configuration	Swagger UI	XML comments, JWT authorization button, response examples
API versioning	API versioning setup	Version 1.0 for Phase 1, URL path versioning (/api/v1/...)
Response standardization	ApiResponse wrapper	Standard format: { success, message, data, errors, timestamp }
Exception handling	Global exception middleware	Handle validation errors, unauthorized, not found, internal errors
Unit tests for controllers	xUnit tests	Test CRUD operations, validation, authorization checks

Completion Criteria for Week 3:

- Full CRUD operations on machines
- Machines can be assigned to work centers
- Status updates are tracked with history
- Swagger UI available at [/swagger](#)

Week 4: IoT Simulation Engine

Day 1-2: Simulation Core Design

Task	Deliverable	Technical Details
Background service	MachineSimulatorService : BackgroundService	Runs continuously, generates simulated machine data
Machine behavior profiles	MachineProfile class	Define cycle patterns: CNC (45s cycle, 2% fault rate), InjectionMolding (30s, 3% fault), Assembly (25s, 1% fault)
Random fault injection	FaultSimulator	Configurable MTBF, random fault generation with realistic durations
State machine logic	MachineStateMachine	Manage transitions: RUN→IDLE→FAULT→SETUP→RUN
Configuration options	SimulationSettings	Enable/disable simulation, speed factor, number of machines

Day 3: Data Generation Engine

Task	Deliverable	Technical Details
Production event generator	EventGenerator	Generate CycleComplete events with timestamps, good counts
Downtime event generator	DowntimeGenerator	Generate realistic downtime based on machine type and reason distribution
Quality event injection	QualitySimulator	Configurable defect rate (0.5% to 5%), random defect type assignment
Shift pattern simulation	ShiftSimulator	Respect shift schedules, lunch breaks, shift change handovers
Performance degradation	DegradationSimulator	Gradual OEE decline over time to simulate wear, or improvement to simulate learning

Day 4: Real-time Data Push

Task	Deliverable	Technical Details
SignalR Hub integration	ProductionHub	Broadcast machine status changes, production counts, alerts
WebSocket connection management	ConnectionManager	Track connected clients, group by role/department
Real-time dashboard data	LiveMachineStatusDto	Push updates every 1-2 seconds for active machines
Heartbeat mechanism	HeartbeatService	Verify simulation is running, restart if stopped

Day 5: Simulation Management & Testing

Task	Deliverable	Technical Details
Simulation control endpoints	SimulationController	Start, Stop, Pause, Resume, Configure simulation
Simulation dashboard	Admin endpoints	View simulation status, adjust parameters live
Performance testing	Load testing script	Simulate 50 machines simultaneously, monitor CPU/memory
Data validation	Validation service	Verify generated data integrity (no gaps, timestamps sequential)

Completion Criteria for Week 4:

- All 5 sample machines generate realistic production data
- OEE values are calculated in real-time
- SignalR pushes updates to connected clients
- Simulation can be controlled via API endpoints

Phase 1 Deliverable Checklist:

- ☒ PostgreSQL database with all tables
 - ☒ JWT authentication working
 - ☒ 8 roles with proper permissions
 - ☒ CRUD APIs for machines and work centers
 - ☒ IoT simulation generating realistic data
 - ☒ SignalR real-time updates
 - ☒ Swagger documentation
 - ☒ Unit tests (minimum 70% coverage for core services)
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Phase 2: Core Production Tracking (Weeks 5-8)

Week 5: Work Order Management

Day 1-2: Work Order CRUD & Business Logic

Task	Deliverable	Technical Details
Work order service	IWorkOrderService	Create, Update, Cancel, Close work orders
Work order validation	WorkOrderValidator	Check material availability, capacity constraints before release
Order numbering strategy	OrderNumberGenerator	Configurable format: WO-{year}-{sequence}
Work order status workflow	WorkOrderStateMachine	Define valid transitions: Draft→Released→InProgress→Paused→Completed→Closed
Work order controller	WorkOrderController	Full CRUD endpoints, authorization for planners only

Day 3-4: Scheduling & Dispatching

Task	Deliverable	Technical Details
Work order assignment	AssignmentService	Auto-assign to best machine based on capability and load
Manual dispatching	DispatchController	Supervisors can assign/reassign work orders to machines
Scheduling algorithm	SchedulingService	FIFO, priority-based, or earliest due date algorithms
Gantt data generation	GanttService	Return formatted data for frontend Gantt chart
Capacity check	CapacityService	Check machine availability for new work orders

Day 5: Work Order Tracking

Task	Deliverable	Technical Details
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Task	Deliverable	Technical Details
Progress tracking	ProgressService	Calculate % complete, estimated remaining time
Work order dashboard endpoints	WorkOrderDashboardDto	Summary by status, overdue orders, completion trends
Work order history	WorkOrderHistoryService	Track all status changes with timestamps and user attribution
Integration with production events	EventWorkOrderLinker	Link production events to work orders automatically

Week 5 Completion:

- Work orders can be created, assigned, and tracked
- Gantt data available for scheduling view
- Real-time progress updates

Week 6-7: Real-time Machine Monitoring Dashboard (Angular + SignalR)

Week 6: Angular Setup & Core Components

Task	Deliverable	Technical Details
Angular project setup	Angular 17+ project	Standalone components, lazy loading modules
Authentication integration	Auth service	JWT storage, login/register components, route guards
SignalR client setup	SignalR service	Manage connection, handle reconnection, typed hub methods
Layout components	Shell component	Sidebar navigation, header with user menu, footer
Machine overview dashboard	MachineDashboardComponent	Grid/card view of all machines with status colors

Week 6-7: Real-time Visualization

Task	Deliverable	Technical Details
Machine status cards	MachineCardComponent	Show name, status, current OEE, production count, live updates via SignalR
Live OEE gauges	OeeGaugeComponent	SVG/CSS circular gauges for Availability, Performance, Quality

Task	Deliverable	Technical Details
Production pace display	PaceDisplayComponent	Current rate vs target rate, green/yellow/red indicators
Machine detail view	MachineDetailComponent	Full machine metrics, recent events, downtime history
Real-time alerts	AlertComponent	Toast notifications for faults, low OEE, threshold breaches

Week 7 Completion:

- Angular app connects to SignalR
- Dashboard shows live machine status
- OEE values update in real-time
- Alerts appear for critical events

Week 8: Production Event Recording & Downtime Tracking

Day 1-2: Production Event Recording

Task	Deliverable	Technical Details
Event recording service	IProductionEventService	RecordCycleComplete, RecordCountUpdate, RecordProductionStart/End
Manual count entry	ManualCountController	Operators submit good/defect counts via tablet interface
Event validation	EventValidator	Ensure counts are sequential, no negative values
Event replay for reporting	EventReplayService	Reconstruct production timeline from events
Bulk event ingestion	BulkEventEndpoint	Accept batch events (for PLC integration later)

Day 3-4: Downtime Tracking

Task	Deliverable	Technical Details
Downtime service	IDowntimeService	StartDowntime, EndDowntime, GetActiveDowntime
Operator downtime entry	DowntimeEntryComponent	Touch-friendly interface for selecting reason codes
Automatic downtime detection	DowntimeDetector	If machine idle > 3 minutes, auto-create downtime record

Task	Deliverable	Technical Details
Downtime reason hierarchy	ReasonTreeService	Parent-child relationships for detailed root cause
Downtime analytics	DowntimeAnalyticsService	Top reasons by machine/shift, duration distribution

Day 5: Integration & Testing

Task	Deliverable	Technical Details
End-to-end workflow test	E2E tests	Operator login → Start production → Record counts → Log downtime → Complete order
Performance test	Load test	100 concurrent operators recording events
Data consistency check	ConsistencyValidator	Verify event sequences, no overlapping downtimes
API integration tests	Integration tests	Full workflow with real database

Week 8 Completion:

- Production events recorded correctly
- Downtime captured with reasons
- Operator tablet interface complete
- End-to-end workflow working

Phase 2 Deliverable Checklist:

-  Work order lifecycle management
-  Gantt scheduling view
-  Angular dashboard with real-time SignalR updates
-  Live OEE gauges and machine status
-  Production event recording
-  Downtime tracking with reason codes
-  Operator tablet interface

Phase 3: Quality Management (Weeks 9-12)

Week 9: Defect Tracking Module

Day 1-2: Defect Data Model & CRUD

Task	Deliverable	Technical Details
Defect entity	Defect, DefectType, DefectCategory	Tables for defect classification hierarchy
Defect service	IDefectService	RecordDefect, GetDefectsByWorkOrder, GetDefectsByMachine
Defect classification	DefectClassificationService	Category: Dimensional → Sub: Length → Defect: Over spec
Defect controller	DefectController	CRUD endpoints, authorization for operators and quality
Defect entry UI	DefectEntryComponent	Operator-friendly defect logging with common defects pre-filled

Day 3-4: Defect Analytics

Task	Deliverable	Technical Details
Pareto analysis	ParetoService	Calculate 80/20 distribution of defect types
Defect trend analysis	DefectTrendService	Daily/weekly/monthly defect rate trends
Defect by shift/operator	OperatorDefectAnalysis	Identify training needs or skill gaps
Scrap calculation	ScrapService	Track scrap vs rework, cost impact calculation
Defect dashboard	DefectDashboardComponent	Pareto charts, trend graphs, heat maps

Day 5: Defect Reporting

Task	Deliverable	Technical Details
Defect report generation	DefectReportService	PDF/Excel export of defect data
Real-time defect alerts	DefectAlertService	Alert when defect rate exceeds threshold (e.g., >3%)
Defect image attachment	FileUploadService	Attach photos of defects (store in Azure Blob/AWS S3)
Search/filter capabilities	DefectSearchService	Search by date, product, machine, defect type

Week 10: Quality Check Management

Day 1-2: Inspection Points

Task	Deliverable	Technical Details
Inspection point entity	InspectionPoint	Define checkpoints in production flow (incoming, in-process, final)
Inspection schedule	InspectionScheduleService	Trigger quality checks at configurable intervals (every N units, every N minutes)
Digital inspection forms	InspectionFormBuilder	Dynamic form builder for quality checks (measurements, pass/fail, dropdowns)
Inspection execution	InspectionService	Record inspection results, auto-calculate pass/fail
SPC data collection	SpcDataService	Collect measurement data for control charts

Day 3-4: Statistical Process Control (SPC)

Task	Deliverable	Technical Details
Control chart calculations	ControlChartService	X-bar, R-chart, P-chart, U-chart calculations
Control limits	ControlLimitService	Calculate UCL/LCL, detect out-of-control conditions
Nelson rules implementation	NelsonRuleService	Detect special cause variation using 8 Nelson rules
Real-time SPC charts	SpcChartComponent	Canvas/Chart.js visualization with live updates
Out-of-control alerts	OocAlertService	Notify quality when control limits violated

Day 5: First Pass Yield (FPY) Analytics

Task	Deliverable	Technical Details
FPY calculation	FpyService	Calculate at station, line, and plant level
Rolled throughput yield	RtyService	Cumulative FPY across multiple operations
Yield waterfall analysis	YieldWaterfallService	Identify yield loss at each operation
FPY dashboard	FpyDashboardComponent	Trends by product, line, shift, operator
Improvement tracking	ImprovementTrackingService	Track FPY improvements after corrective actions

Week 11-12: Digital Quality Forms & Workflow Integration

Week 11: Dynamic Form System

Task	Deliverable	Technical Details
Form builder backend	FormBuilderService	CRUD for form templates, field types (text, number, checkbox, dropdown, date, file)
Form rendering engine	FormRenderService	Render JSON form definition as HTML/Reactive Form
Form submission service	FormSubmissionService	Store submissions in database, link to production events
Form validation	FormValidationService	Client-side and server-side validation based on rules
Form response dashboard	FormResponseComponent	View submitted forms, export to PDF/Excel

Week 12: Quality Workflow Integration

Task	Deliverable	Technical Details
Quality gate workflow	QualityGateService	Block production if quality checks fail
Sampling plans	SamplingPlanService	AQL-based sampling (ANSI/ASQ Z1.4)
Quality holds	QualityHoldService	Place work orders on hold, require manager approval to release
Rework tracking	ReworkService	Track rework orders, rework time, rework success rate
Integration with work orders	QualityWorkOrderLink	Link quality issues to specific work orders for traceability

Phase 3 Deliverable Checklist:

- ✔ Defect tracking with classification hierarchy
- ✔ Pareto and trend analysis
- ✔ Digital inspection forms
- ✔ SPC charts with Nelson rules
- ✔ FPY and RTY calculations
- ✔ Quality hold workflow

Phase 4: Analytics & Reporting (Weeks 13-16)

Week 13-14: OEE Calculation & Dashboards

Week 13: OEE Core Engine

Task	Deliverable	Technical Details
OEE calculation service	OeeService	Calculate Availability, Performance, Quality, OEE
Availability calculation	AvailabilityService	$(\text{Run Time} / \text{Planned Production Time}) \times 100$
Performance calculation	PerformanceService	$(\text{Ideal Cycle Time} \times \text{Total Parts}) / \text{Run Time} \times 100$
Quality calculation	QualityService	$(\text{Good Parts} / \text{Total Parts}) \times 100$
Time classifications	TimeClassificationService	Planned production time, run time, downtime, idle time
OEE by time period	OeeTimeSeriesService	Hourly, shift, daily, weekly, monthly OEE

Week 14: OEE Dashboards & Loss Analysis

Task	Deliverable	Technical Details
Real-time OEE dashboard	OeeDashboardComponent	Live OEE with 6 big losses breakdown
Six big losses visualization	BigLossesComponent	Breakdown by: Breakdowns, Setup, Idling, Slow Cycles, Defects, Rework
OEE by machine comparison	MachineComparisonComponent	Bar chart comparing OEE across machines
Loss tree drilldown	LossTreeComponent	Hierarchical view of losses: Availability → Downtime → Mechanical → Bearing Failure
OEE target tracking	TargetTrackingComponent	Actual vs target, trend arrow, days since target achieved
Export OEE reports	OeeReportExport	PDF/Excel with charts and raw data

Week 15: Production Analytics & Trend Analysis

Task	Deliverable	Technical Details
Throughput analysis	ThroughputService	Units per hour, per shift, trend analysis

Task	Deliverable	Technical Details
Bottleneck detection	BottleneckService	Identify constraint operations using cycle time and queue length
Cycle time analysis	CycleTimeService	Actual vs standard, Cpk calculation, distribution analysis
MTBF/MTTR calculation	MaintenanceAnalyticsService	Calculate per machine, per failure type
Trend forecasting	TrendForecastService	Linear regression for OEE, throughput, quality trends
Heat maps	HeatMapService	Time-of-day/dow performance heatmaps
Correlation analysis	CorrelationService	Find correlations (e.g., temperature vs defect rate)

Week 16: Report Generation & Export

Task	Deliverable	Technical Details
Report engine architecture	ReportEngineService	Modular report generation with templates
Standard report templates	Predefined reports	Shift summary, daily production, weekly OEE, monthly quality, downtime analysis
Scheduled reports	ScheduledReportService	Email reports daily/weekly/monthly using Hangfire/Quartz
Custom report builder	ReportBuilderComponent	Drag-and-drop fields, filters, date ranges, aggregations
Export formats	ExportService	CSV, Excel (EPPlus), PDF (jsPDF/iTextSharp)
Report scheduling UI	ReportScheduleComponent	Configure frequency, recipients, format, delivery method
Report history	ReportHistoryService	Store generated reports, allow re-download
Dashboard widgets	WidgetService	Configurable widgets for personalized dashboards

Phase 4 Deliverable Checklist:

- ✔ Accurate OEE calculation with six big losses
- ✔ Real-time OEE dashboard
- ✔ MTBF/MTTR analytics
- ✔ Bottleneck detection
- ✔ Comprehensive reporting engine

- ✔ Scheduled report delivery

Phase 5: Advanced Features (Weeks 17-20)

Week 17: CAPA Workflows (Corrective and Preventive Action)

Task	Deliverable	Technical Details
CAPA entity	Capa, CapaTask, CapaAttachment	Tables for CAPA records, tasks, evidence
CAPA workflow engine	CapaWorkflowService	States: Identify → Evaluate → Plan → Implement → Verify → Close
Escalation rules	EscalationService	Auto-escalate if defect rate >5% or same defect repeats 3+ times
Root cause analysis	RootCauseService	5-Why analysis, Fishbone diagram data structure
Effectiveness verification	EffectivenessService	Track metrics after CAPA implementation
CAPA dashboard	CapaDashboardComponent	Open CAPAs by age, overdue tasks, effectiveness rates
Notification system	NotificationService	Email/in-app notifications for assignments and due dates
CAPA approval workflow	ApprovalWorkflowService	Multi-level approval for major CAPAs

Week 18: Predictive Analytics Foundation

Task	Deliverable	Technical Details
Data preprocessing pipeline	DataPrepService	Clean and prepare historical data for ML models
Anomaly detection	AnomalyDetectionService	Statistical (Z-score, IQR) and ML-based (Isolation Forest)
Predictive maintenance model	PredictiveMaintenanceService	ML.NET binary classification for failure prediction
Quality prediction	QualityPredictionService	Predict defect probability based on machine parameters

Task	Deliverable	Technical Details
OEE forecasting	OeeForecastService	Time series forecasting (ARIMA, Prophet)
ML model management	ModelManagementService	Train, save, load, version ML models
Prediction API endpoints	PredictionController	Real-time predictions with confidence scores
Model retraining scheduler	RetrainingService	Weekly retraining with new data

Week 19: Mobile-Responsive Design

Task	Deliverable	Technical Details
Responsive CSS framework	TailwindCSS/Bootstrap configuration	Mobile-first breakpoints, touch-friendly targets
Operator PWA	Progressive Web App	Installable, offline support for critical functions
Mobile dashboard	MobileDashboardComponent	Simplified view for phones, quick actions
Touch-optimized controls	TouchComponents	Large buttons, swipe gestures, voice input (optional)
QR code scanning	QRScannerService	Scan work order QR codes, machine IDs
Mobile push notifications	PushNotificationService	Alert operators on mobile when machine stops
Offline data sync	SyncService	IndexedDB for offline data, sync when online
Role-specific mobile views	RoleViews	Different mobile layouts for operators vs supervisors vs maintenance

Week 20: System Administration Tools

Task	Deliverable	Technical Details
Admin dashboard	AdminDashboardComponent	System health, user activity, audit log viewer
Audit trail viewer	AuditLogComponent	Searchable, filterable history of all data changes
Configuration management	ConfigService	Dynamic configuration without redeployment

Task	Deliverable	Technical Details
Database maintenance	DbMaintenanceService	Archive old data, rebuild indexes, update statistics
Health checks	HealthCheckEndpoint	/health endpoint for load balancers and monitoring
Performance monitoring	MetricsService	Track API response times, database query performance
Backup/restore interface	BackupService	Schedule database backups, one-click restore
User activity monitoring	UserActivityService	See who's logged in, last activity, force logout
Role management UI	RoleManagementComponent	Create/Edit roles, assign permissions dynamically
System logs viewer	LogViewerComponent	View Serilog logs from database with filtering
API usage analytics	ApiAnalyticsService	Most used endpoints, by user, response times
Bulk data operations	BulkOperationService	Bulk update machines, work orders, users

Phase 5 Deliverable Checklist:

- ✔ Complete CAPA workflow with escalation
- ✔ Predictive maintenance alerts
- ✔ Mobile PWA for operators
- ✔ System administration dashboard
- ✔ Comprehensive audit trails
- ✔ Health monitoring and backup/restore

Cross-Phase Integration Points

SignalR Hubs Overview

Hub Name	Purpose	Clients
ProductionHub	Live machine status, production counts	All authenticated users
AlertHub	Real-time alerts and notifications	Supervisors, managers
QualityHub	Quality check notifications	Quality inspectors

Hub Name	Purpose	Clients
MaintenanceHub	Fault alerts, repair status	Maintenance team

Background Services Summary

Service	Phase	Interval	Purpose
MachineSimulatorService	1	Continuous	Generate IoT data
OeeCalculationService	4	Every 5 min	Pre-calculate OEE aggregates
ReportGenerationService	4	Scheduled	Generate and email reports
ModelTrainingService	5	Weekly	Retrain ML models
DataArchivalService	5	Daily	Archive old production events

API Versioning Strategy

Version	Phase	Status
v1.0	1-3	Current
v2.0	4-5	Future (backward compatible)

Technology Stack Summary

Layer	Technology	Version
Backend	ASP.NET Core	8.0
Database	PostgreSQL	15+
ORM	Entity Framework Core	8.0
Auth	JWT + ASP.NET Core Identity	8.0
Real-time	SignalR	8.0
Frontend	Angular	17+
UI Library	Angular Material / PrimeNG	Latest
Charts	Chart.js / ECharts	Latest
ML.NET	Microsoft.ML	3.0
Logging	Serilog	Latest

Layer	Technology	Version
Testing	xUnit, Moq	Latest

Success Metrics Per Phase

Phase	Key Metrics	Target
Phase 1	API response time, Database migration success	<100ms, 100%
Phase 2	Dashboard load time, SignalR latency	<2s, <500ms
Phase 3	Defect entry time, SPC alert accuracy	<10s, >95%
Phase 4	OEE accuracy vs manual, Report generation	>95%, <5s
Phase 5	Prediction accuracy, PWA load time	>80%, <3s

This detailed breakdown provides a complete roadmap for all 20 weeks of development across the 5 phases. Each task is specific, measurable, and includes the technical implementation details needed for your development team to execute successfully.